Runic Arcana Game Design Document

Made by Arcana Team

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RMIT UNIVERSITY

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Project Overview

Team

Doan Phu Khang - Project Lead

- ★ In charge of project direction, game design, and coding.

Trieu Khang Vy - Designer

- ★ In charge of map, art, UI, and puzzle design.

Nguyen Duc Son Hai - Artist

- ★ In charge of art designing, animating, and documenting.

Pitch

In the realm of magic, you're just a lowly mage minion, living a dull existence serving a harsh General. Each day is the same—collecting loot, carrying out orders, enduring the General's rude commands. But one day, everything changed.

During a routine mission, you stumble into a hidden jungle chamber and discover a Magical Tomb filled with mysterious runes. In a moment of curiosity, you trigger a powerful Thunder spell, causing chaos and destroying most of the runes.

Realizing your mistake, you return home to report the incident to the General, only to find him trying to steal the magical book. Feeling betrayed, you escape with the tome, embarking on a journey to find the scattered runes and uncover the secrets they hold.

Setting

The game takes place in a fantasy world with magic, runes, and monsters. Moreover, the theme was also inspired by ancient Scandinavian lands from the Vikings' mythical stories.

Genre

2D top-down adventure & action game.

Game Topic

Main topic: Education

Cognitive Learning

Runic Arcana focuses on educating players in developing cognitive learning skills. But what exactly does cognitive learning mean?

Cognitive learning is a broad theory that explains thinking and understanding as the foundation for learning. This theory focuses on the inner processes of the mind and how these processes influence the way we acquire knowledge.

Key elements of cognitive learning include:

1. **Active Learning:** Cognitive learning emphasizes the learner's active role. Rather than passively receiving information, learners engage with the material through questioning, exploring, and applying knowledge.

2. **Mental Processes**: It focuses on the internal mental processes such as thinking, memory, knowing, and problem-solving. Understanding how these processes work is crucial to improving learning and retention.

3. **Schema Theory:** This concept, introduced by Jean Piaget, involves the idea that knowledge is organized into units or schemas. As people learn, they either assimilate new information into existing schemas or accommodate by changing schemas to include new information.

4. **Information Processing:** This aspect of cognitive learning examines how people process information, from taking in information (encoding), storing it, and retrieving it when needed.

Important figures in Cognitive Learning research:

Jean Piaget: Known for his theory of cognitive development, Piaget's work highlights how children develop cognitive abilities through stages.

Lev Vygotsky: He emphasized the social context of learning and introduced the concept of the Zone of Proximal Development (ZPD), which describes tasks a learner can perform with guidance but not yet independently.

Jerome Bruner: He developed the theory of discovery learning and emphasized the importance of structure in learning. Bruner believed that learning is an active process where learners construct new ideas based on their current and past knowledge.

Cognitive Learning in Games

Cognitive learning in games leverages the principles of cognitive learning theory to enhance educational experiences and skill acquisition through gameplay. This approach utilizes the interactive and immersive nature of games to promote active learning, problem-solving, and critical thinking.

Examples:

1. **Educational Games**: These are explicitly designed to teach specific skills or knowledge:

- **“Minecraft: Education Edition”:** Encourages creativity, collaboration, and problem-solving through building and exploring.
- **“Kerbal Space Program”:** Teaches physics, engineering, and space exploration principles through designing and launching spacecraft.

2. **Puzzle Games:** These games challenge players' cognitive abilities by requiring them to solve puzzles, which can enhance logical reasoning and pattern recognition:

- **“Portal”:** Involves solving physics-based puzzles using a portal gun, encouraging spatial reasoning and problem-solving.
- **“The Witness":** Features complex puzzles that require players to observe patterns and use deductive reasoning.

3. **Strategy Games:** These games involve planning, resource management, and strategic thinking:

- "**Civilization**": Players manage resources, develop civilizations, and make strategic decisions to expand their empire.
- "**Starcraft**": A real-time strategy game that requires quick thinking, resource management, and strategic planning

4. **Simulations:** These games simulate real-world activities, allowing players to apply theoretical knowledge practically:

- "**SimCity**": Teaches urban planning, resource management, and the impact of various decisions on a city.
- **“Flight Simulator**": Provides a realistic flying experience, teaching principles of aviation and navigation.

Benefits of Cognitive Learning in Games:

- **Enhanced Engagement**: The interactive nature of games keeps learners engaged and motivated.
- **Safe Learning Environment:** Games provide a safe space to experiment and learn from mistakes without real-world consequences.

- Adaptive Learning: Games can adapt to the learner's skill level, providing personalized learning experiences.
- Collaboration: Many games encourage teamwork and cooperation, enhancing social learning and communication skills.
- Critical Thinking: Games often require players to think critically and make decisions based on incomplete information.

Implication of Runic Arcana:

By experiencing the game, players will get the chance to learn the skills of precise drawing and fast reaction time. To conquer the massive waves of monsters, players need to master the seamless transition between casting powerful spells and gracefully dodging enemy attacks.

Secondly, players will also be rewarded if they spend time exploring the game's world, and the prize for their curiosity is powerful spells that are exclusive from the main pathway. With this, Runic Arcana teaches players to think outside the box and greatly inspires players to expand their curiosity toward things around them.

Moreover, the runic alphabet in the game was inspired by the ancient Scandinavian runes, also known as fuþark or futhark. This adds a touch of historical knowledge to the game. Solving the puzzle would also greatly increase players' knowledge of the runic alphabet, which could potentially become an inspiration source for their future projects.

Target Platform

PC, but could potentially be ported onto devices with touch screens (Phones, tablets, etc...)

Target Audience

Casual players with great interest in magical worlds.

Gameplay

Players explore the world to fight enemies and collect runes.

There will be a guideline on where players should go. However, they will be rewarded for exploring hidden pathways.

A boss fight awaits the players at the end of their journey. Defeating the boss is the end-game condition.

Unique Selling Point

Casting spells by drawing is the unique selling point of our game. This innovative mechanic also delivers a cognitive learning experience, requiring players to remember and replicate spell shapes to cast them. It's a fun, intuitive, and groundbreaking way to engage and challenge players, making every spell a rewarding and satisfying achievement.

Mechanics

Movement Mechanics:

- Moving with standard WASD keys.

Battle Mechanics:

- Cast spells by drawing using a mouse.
- Spell Variation: Attack, Defense (Fireball, Ice spikes, Shield, etc...).

Obstacle Types:

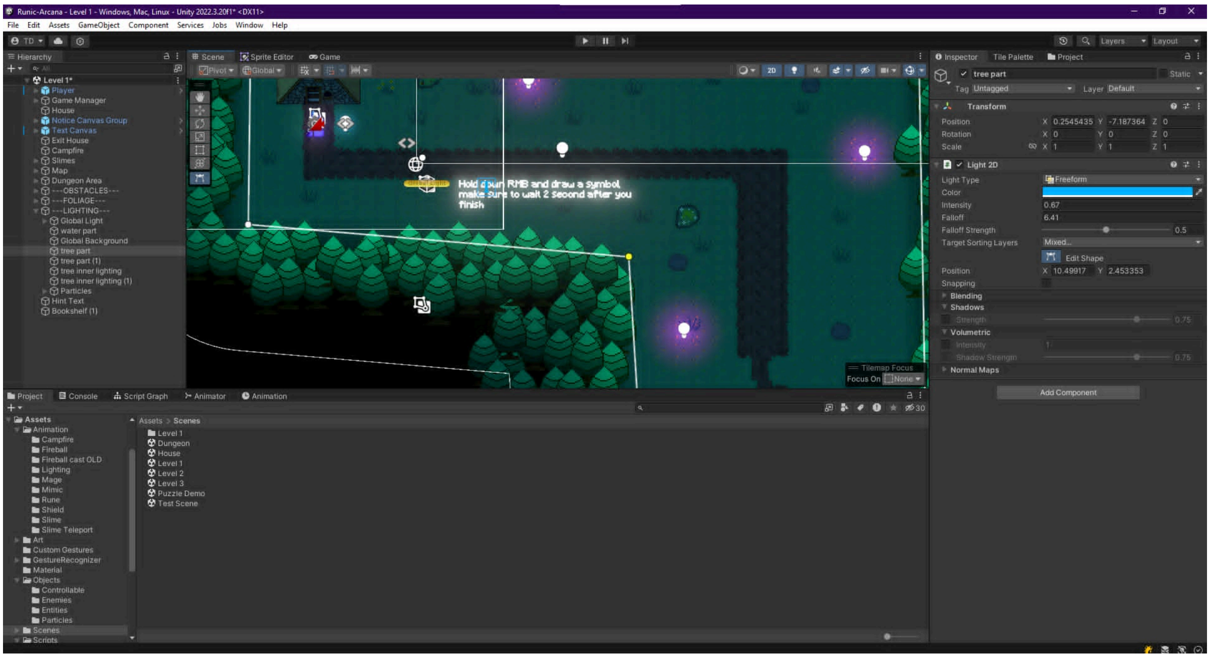
- Riddle Puzzle: Reading + Analyzing.
- Monsters (Blue slime, Skeleton, etc...).
- Boss Fight (End-game condition).

Power-Ups:

- Explore the world for more runes (More runes = more spells).
- Puzzle rewards the player.

Working Process

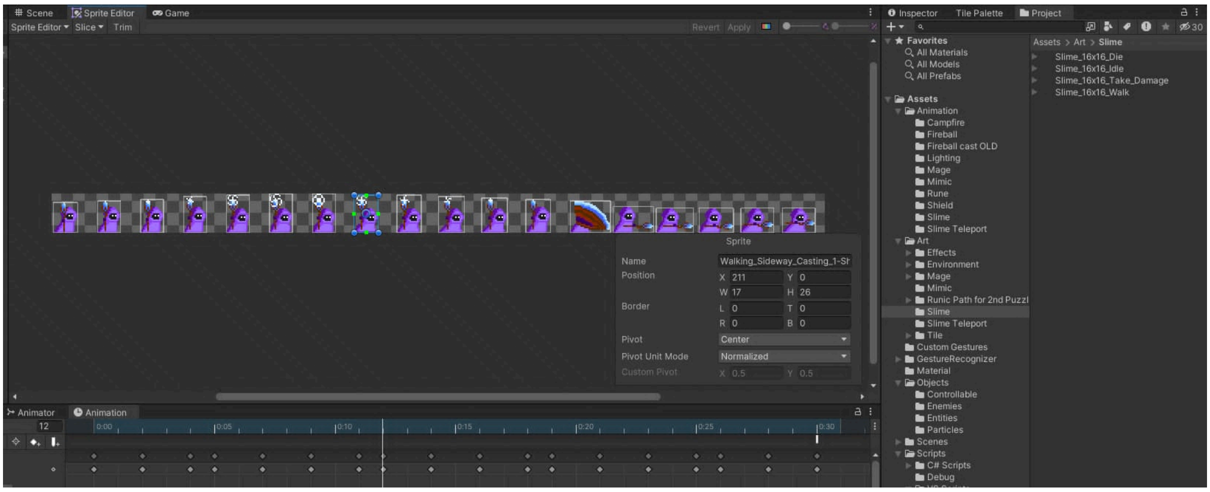
Level building



The overall map was created first, before being refined later by adding glowing guidelines, grass, roads, etc… to make it more appealing.

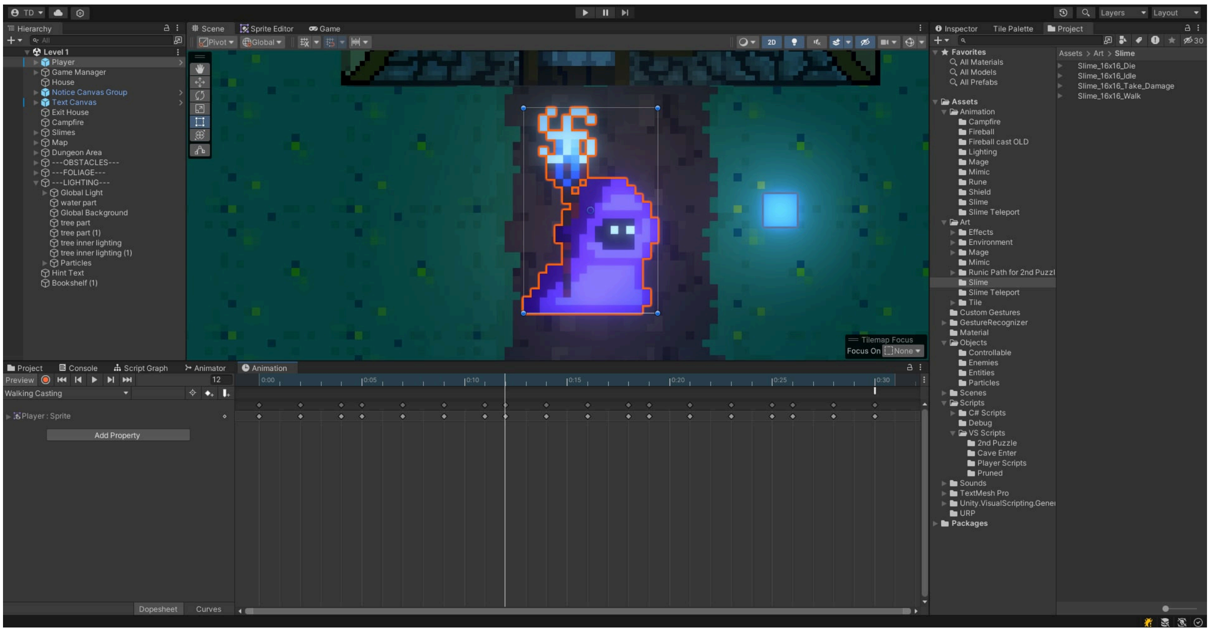
After that, the borderline was added (as another layer), making it much easier to work with rather than placing everything in the same tileset.

Inserting sprite sheets



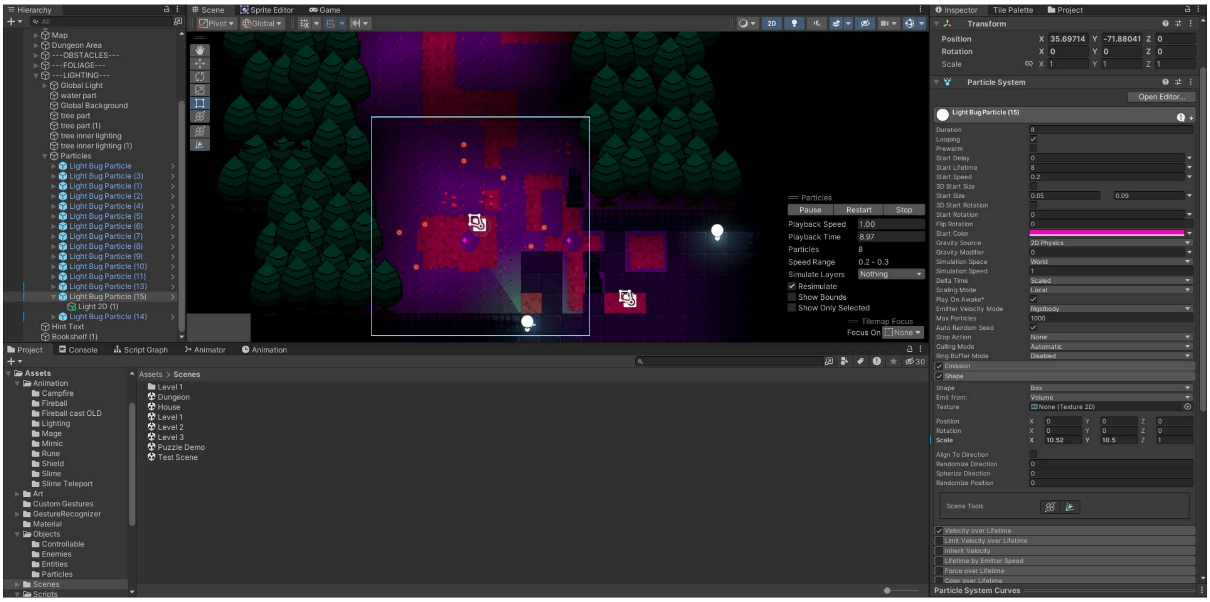
Sprite sheets were manually sliced.

Character animation



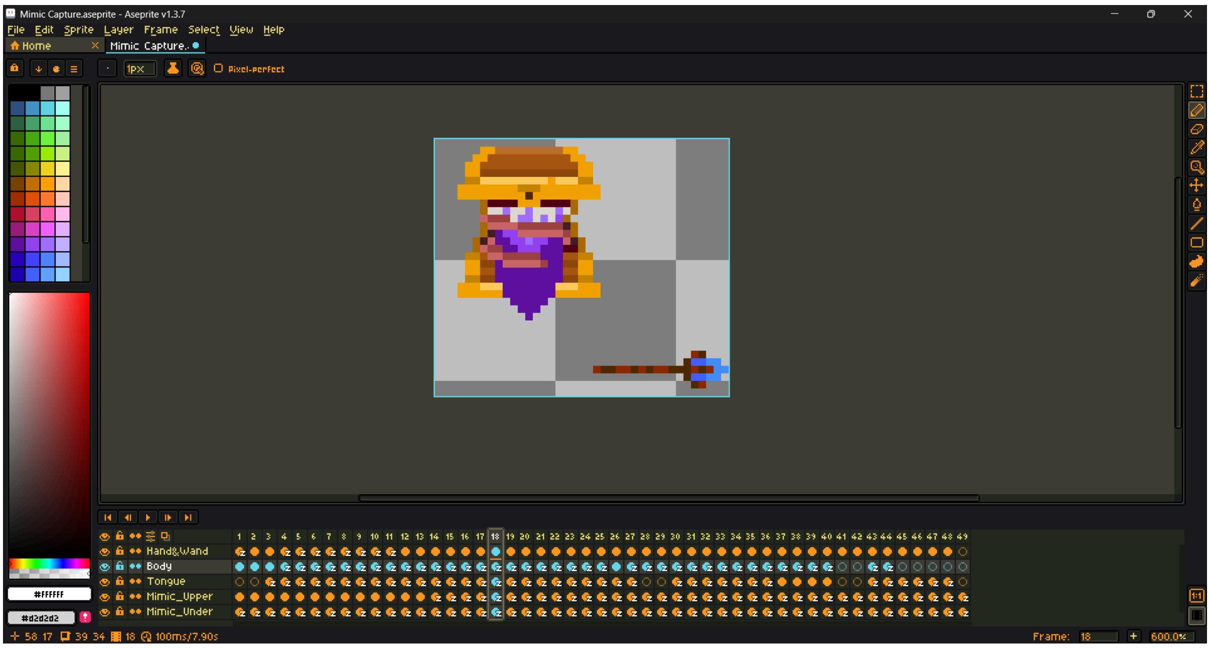
Each key frames were added in with the corresponding timing to make the animation smooth and beautiful.

Lighting



Lighting was added to make things stand out (and for the aesthetic aspect too). With the highlighted guidelines, players can easily figure out the way to the finishing line.

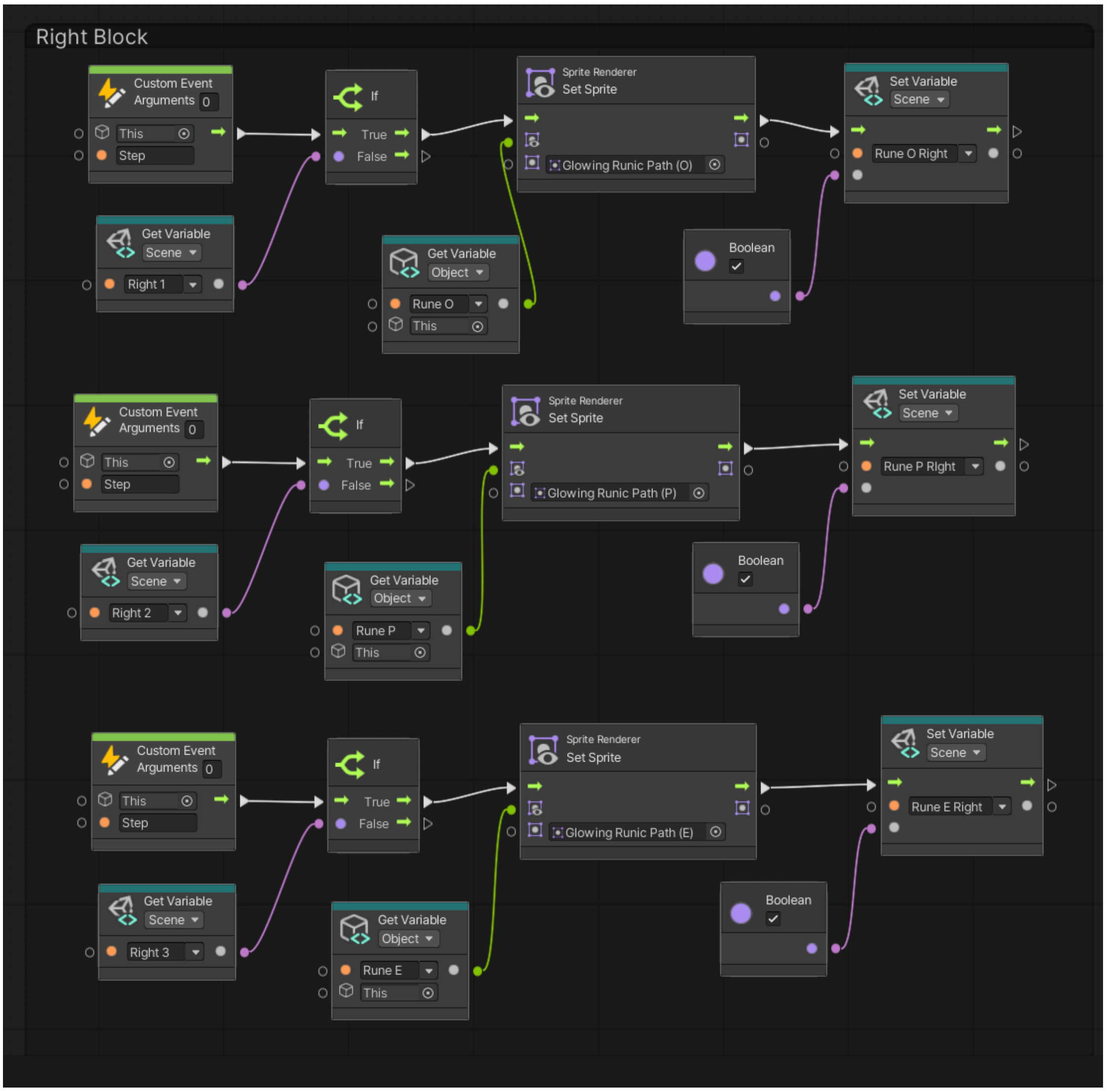
Drawing / Animating



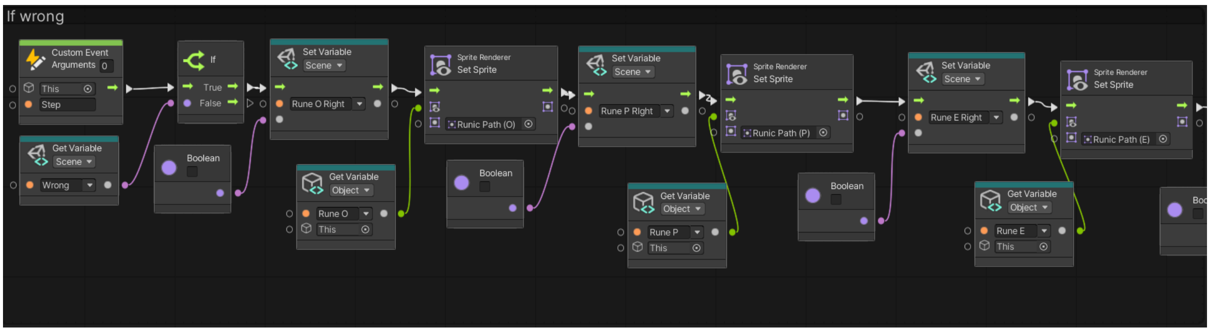
Every sprite was drawn and animated on Aseprite.

Visual Scripting Explanation

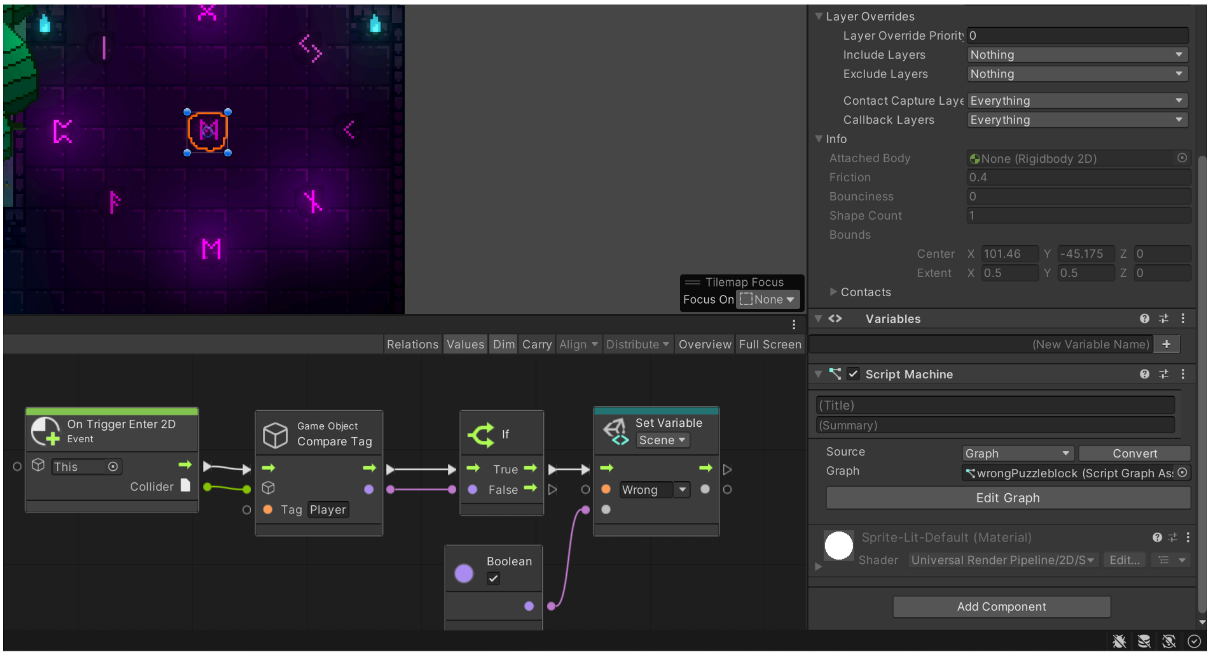
In the game's puzzle, if players choose the correct rune, the gate will open, allowing them to continue the gameplay. To choose the right rune, players need to search for hints.



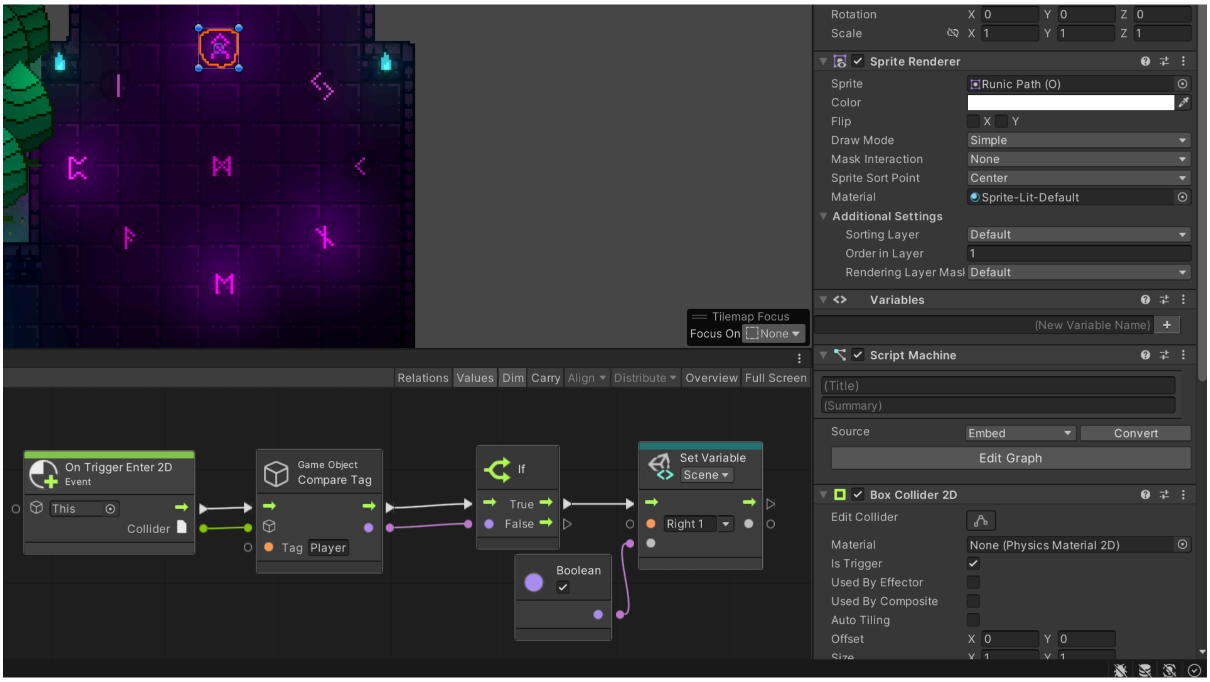
When players choose the correct rune, this VS code will change the sprite of that rune to indicate it is accurate. Then, it will set the 'Right Rune' variable to true. If all runes are correct, the gate bar will be removed, allowing the player to access a new area to explore.



However, with this VS code, if the player chooses an incorrect rune, the puzzle will reset, and they will have to solve it again from the beginning. This serves as a punishment for players who try to bypass the hints by randomly selecting runes

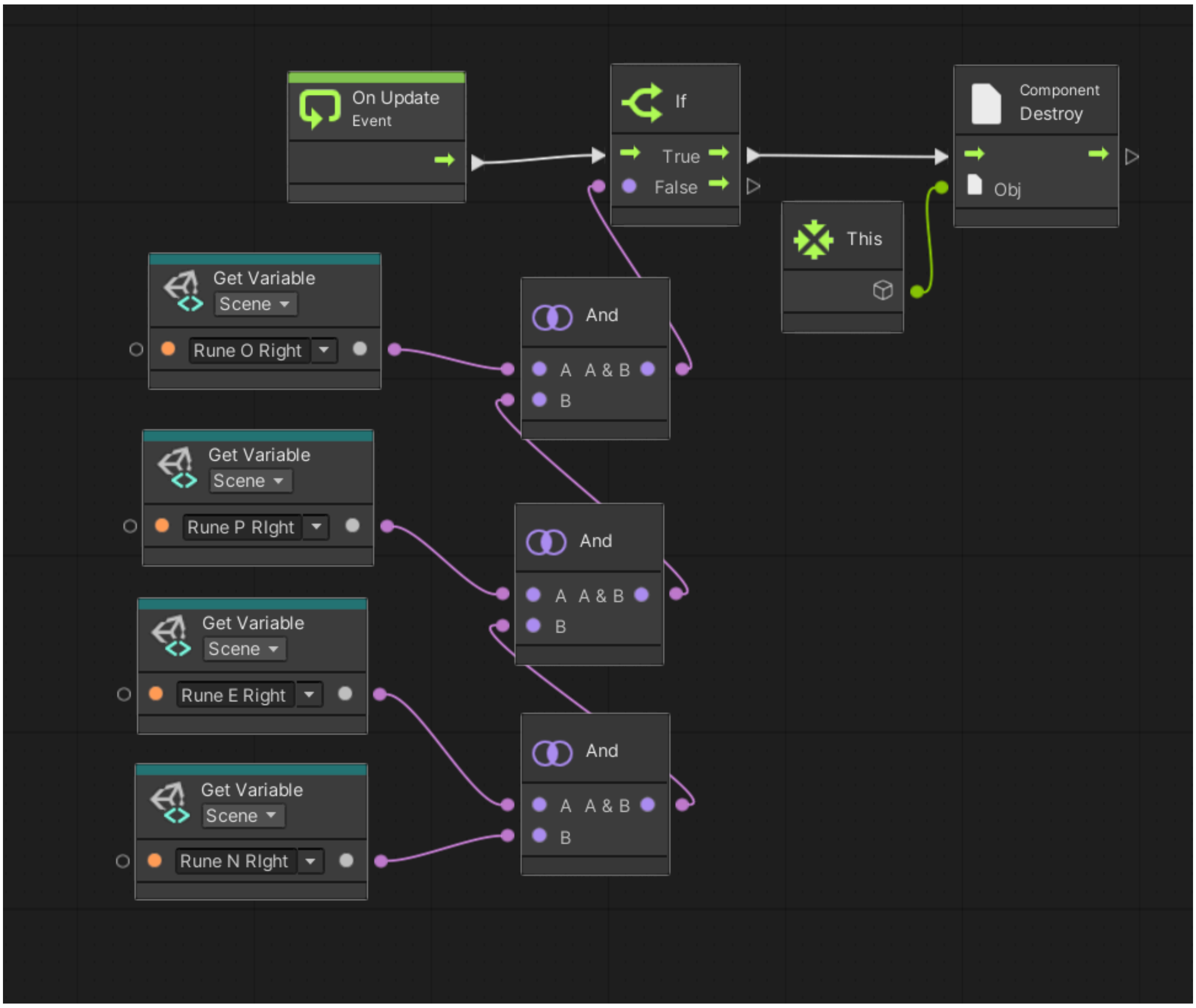


The image above shows how the incorrect runes look. When the player steps on one, it will set the 'Wrong' variable to true. If they then press Space to choose it, the puzzle will reset. Therefore, always look for hints before solving this puzzle, or it might take many attempts to complete it correctly.



This is the correct rune code. For the specific puzzle shown in the image above, there are 4 correct runes. Therefore, there will be 4 'Right' variables. When the player steps on a correct rune and presses Space to choose it, it will change the sprite and set the 'Right 1' variable to true. There will still be three more correct runes to find to solve this.

Finally, the most important and simple code is to remove the gate bar when all the correct runes have been chosen.



With this VS code, the player can choose the runes in any order. It doesn't matter if the player chooses the E rune first and the O rune later; the gate will still open as long as they choose all the correct runes.